

Project Title
Digital Library Search System

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

in

A.Y. 2026-2027

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ABSTRACT

Modern libraries hold vast and continuously growing collections of books, journals, magazines, and research papers, making manual, linear search increasingly impractical for readers who need accurate results within seconds. This project presents the design and implementation of a Digital Library Search System, a repository-driven application that allows users to locate resources efficiently using titles, authors, ISBNs, categories, and keywords. The system is architected around core Data Structures and Algorithms principles: records are organized using indexed and hash-based structures to support near-constant-time lookups, while string-matching techniques such as pattern-search and prefix-based comparison enable fast and reliable keyword search across large text collections. The underlying repository is modelled so that every book, journal, or research paper is treated as a searchable record, with each attribute indexed independently so that a query on any single field, or a combination of fields, resolves without scanning the entire collection.

Particular attention is given to algorithmic efficiency and scalability, ensuring that search performance remains consistent as the size of the repository increases and that response times do not degrade as new records are added. The initial implementation establishes a robust foundation with exact-match search capabilities across all indexed fields, along with basic error handling and input validation to keep the system reliable under everyday use. This foundation is deliberately structured for extension toward approximate and intelligent search features such as fuzzy matching for misspelled queries, weighted relevance ranking, and category-based filtering, comparable to those found in contemporary digital library platforms. By combining sound algorithmic design with practical software engineering, this project demonstrates how classical data structures and search algorithms can be applied to solve a real-world information-retrieval problem, offering both a functional tool for everyday library use and a case study in building scalable, efficient, and extensible search systems.

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